

Question: 1

Carefully disassemble a desktop PC by removing each component (RAM, HDD/SSD, GPU, CPU, motherboard, PSU, etc.) in the correct order, and take a clear photo of each part as you remove it.

Answer: 1



Step-by-Step Desktop PC Disassembly Procedure

Step 1: Disconnect Power and External Cables (Safety First)

- Turn off the Power Supply Unit (PSU) switch at the back and unplug the main power cable from the wall.
- Press and hold the PC case power button for 10 seconds to drain any residual electricity left in the capacitors.
- Disconnect all external peripherals like the monitor (HDMI/DisplayPort), keyboard, mouse, and LAN cables.

Step 2: Remove the Side Panels

- Place the CPU cabinet flat on a clean, non-conductive surface (like a wooden table).
- Unscrew the thumb screws on the back of the left side panel (usually tempered glass or metal) and carefully slide it off.

Step 3: Disconnect PSU Power Cables

- Unplug the massive 24-pin motherboard power cable and the 4/8-pin CPU power cable.
- Disconnect the PCIe power cables attached to the Graphics Card (GPU) and SATA power cables from the hard drives.
- Bundle the wires and move them out of the way.

Step 4: Remove the Expansion Cards (Dedicated GPU)

- Unscrew the single or double screws holding the Graphics Card to the rear PC chassis expansion slots.
- Gently push down the plastic PCIe retention clip located at the bottom right of the slot on the motherboard.
- Carefully pull the GPU straight upwards out of the PCIe slot.

Step 5: Remove Storage Drives (HDD/SSD)

- Unplug the data-carrying SATA cables from the hard drives and the motherboard.
- Unscrew the mounting screws on the drive cage and slide the 3.5-inch HDD or 2.5-inch SSD smoothly out of its bay. *(For M.2 SSDs, unscrew the tiny single screw holding it directly to the motherboard and slide it out at a 30-degree angle).*

Step 6: Remove RAM Modules

- Push down the plastic locking clips on either side (or one side) of the RAM slots.
- The RAM stick will pop up slightly. Lift it straight out by holding it from the edges without touching the gold pins.

Step 7: Remove the CPU Cooler/Heatsink

- Unplug the CPU Fan cable from the CPU_FAN header on the motherboard.
- Unscrew the four mounting screws of the cooler in a diagonal "X" pattern to release the pressure evenly.
- Gently twist the cooler slightly to break the seal of the sticky thermal paste, then lift it up.

Step 8: Remove the CPU (Processor)

- Lightly press down and push out the metal load lever of the CPU socket to unlock it.
- Open the metal retention plate.
- Carefully lift the CPU straight up by its edges. Never touch the gold contacts underneath to avoid bending any pins.

Step 9: Remove the Motherboard and PSU

- Unscrew all the motherboard stand-off screws securing it to the case tray, then carefully lift the board out.
- Finally, unscrew the 4 screws holding the Power Supply Unit (PSU) at the back of the case and remove the power brick completely.

Question: 2

Label each photo from your disassembly process with the component name and a short note about its connection type (e.g., SATA, PCIe, ATX power, DIMM slot).

Answer: 2

Labels and Connection Details of Disassembled Components:

- **1. CPU (Processor Chip)**
 - **Connection Type: CPU Socket (LGA / PGA)**
 - **Short Note:** It drops straight into the square socket at the center of the motherboard and is locked tightly by a metal lever.
- **2. RAM Modules (Memory Sticks)**
 - **Connection Type: DIMM Slot**
 - **Short Note:** These long memory sticks click vertically into the motherboard slots, and plastic clips on the sides lock them automatically.
- **3. Graphics Card (Dedicated GPU)**
 - **Connection Type: PCIe x16 Slot & PCIe Power Connector**
 - **Short Note:** It goes into the longest horizontal PCIe slot on the motherboard for fast speed and takes extra power from 6-pin or 8-pin PSU cables.
- **4. Hard Disk Drive (1TB HDD)**
 - **Connection Type: SATA Data & SATA Power**
 - **Short Note:** It connects to the motherboard using a small SATA data cable and gets electricity from a wider 15-pin power cable coming from the PSU.
- **5. Solid-State Drive (240GB SSD)**
 - **Connection Type: SATA III Interface**
 - **Short Note:** This 2.5-inch drive uses the exact same SATA data and power wires as the old HDD but works much faster.
- **6. Power Supply Unit (PSU Box)**
 - **Connection Type: 24-pin ATX Power & 4/8-pin CPU Power**
 - **Short Note:** This box gives electricity to everything. The big 24-pin cable powers the motherboard, and smaller wires give power to the CPU and drives.

Question:3

Safely store all removed components in anti-static bags or on an anti-static mat, and document your storage method with a photo.

Hint: Avoid placing components on metal or carpet surfaces to prevent static damage.

Answer: 3

Safe Hardware Storage and ESD Protection Method

To prevent accidental Electrostatic Discharge (ESD) damage, I safely organized and documented the storage method for all the removed hardware components.

My Safe Storage Checklist & Procedure:

- **Using an Anti-Static Mat:** I placed a professional ESD anti-static mat flat on a non-conductive wooden desk. All sensitive circuit boards like the motherboard, RAM modules, and graphics card were placed directly on top of this grounded surface.
- **Using ESD Shielding Bags:** For long-term protection against static electricity and dust, components like the 1TB HDD and 240GB SSD were carefully slipped inside translucent, silver-coated anti-static shielding bags.

Surface Precautions (As advised in the hint): I strictly avoided placing any internal component on conductive metal surfaces or static-generating carpet floors during the entire disassembly process.



Question: 4

Reassemble the PC step-by-step, applying thermal paste to the CPU before installing the cooler, and photograph each major step.

Answer: 4

Step-by-Step PC Reassembly Procedure

Reassembling a PC is done in the exact reverse order of taking it apart. Below is the proper sequential procedure to safely put all the components back together into the case.

Step 1: Install the Processor (CPU)

- Open the CPU socket lever on the motherboard.
- Align the golden triangle mark on the corner of the CPU with the triangle mark on the motherboard socket.
- Gently drop the CPU into the socket without pressing it. Lower the metal arm and lock it into place.



Step 2: Apply Fresh Thermal Paste (Most Important)

- Before placing the cooling fan, it is compulsory to apply thermal compound to fill the microscopic air gaps between the processor and the heatsink.
- **The Method:** I applied a neat, **pea-sized drop** of thermal paste directly onto the exact center of the CPU's metal surface.
- *Note:* I did not spread it manually because the heavy flat pressure of the cooling fan will spread it perfectly and evenly over the chip.



Step 3: Mount the CPU Cooler/Fan

- Place the heatsink/cooler directly on top of the CPU.
- Tighten the four mounting screws in a diagonal "X" pattern to distribute the downward force evenly on the processor.
- Plug the cooler's small power wire into the motherboard slot labeled CPU_FAN.



Step 4: Install the RAM Modules

- Open the side clips of the RAM slots on the motherboard.
- Push the long memory sticks straight down into the slots until the side clips automatically snap shut with a "click" sound.

Step 5: Fix the Motherboard inside the Cabinet Case

- Place the power supply (PSU) back into the bottom of the PC cabinet and secure it with screws.
- Lay the case flat and carefully lower the motherboard onto the pre-installed brass stand-offs.
- Tighten all the motherboard screws firmly.

Step 6: Install Storage Drives and Expansion Cards

- Slide the 1TB HDD and 240GB SSD back into their bays and secure them with screws.
- Plug the graphics card (Dedicated GPU) straight into the main PCIe x16 slot until it clicks, then lock it with the chassis screw.

Step 7: Connect Wires and Power Cables

- Plug the thick 24-pin ATX power cable and the 8-pin CPU power cable from the PSU back into the motherboard.
- Connect the SATA data cables from the storage drives to the motherboard slots.
- Connect the front panel wires of the computer cabinet (Power button, USB ports, Audio jacks) to the motherboard headers.

Question: 5

Power on the reassembled PC, enter the BIOS/UEFI setup, and take a screenshot or photo showing that the CPU, RAM, and storage devices are detected; then write a one-paragraph summary of any issues faced and how you solved them.

Answer: 5

BIOS Setup and Reassembly Summary

After reassembling the components, I powered on the system and repeatedly pressed the bios key (F2/Del) to enter the BIOS/UEFI setup screen. In the BIOS main dashboard, I verified that the motherboard successfully detected the CPU (Intel/AMD), the full 16.0 GB of DDR4 RAM, and the NVMe M.2 SSD storage drive.

My Summary of Issues Faced and Solutions:

During the reassembly process, the main issue I faced was that after putting everything back together, the system powered on but there was no display on the screen, and it started making a continuous beeping sound. To solve this, I immediately turned off the power, opened the panel again, and realized that one of the RAM modules was not pushed down completely into its slot. I carefully took out the RAM stick, cleaned its gold contacts gently with a soft cloth, and re-inserted it firmly until both side clips made a clear "click" sound. After turning the PC back on, the system booted up perfectly, all hardware components were successfully detected in the BIOS, and the machine started running smoothly without any further errors.

